

1. Role

Purpose: Stores all available system roles used for role-based access control.

Column	Data Type	Constraints	Description
Id	INT	PK	Unique role identifier
Name	NVARCHAR(50)	NOT NULL	Role name (Admin, Manager, IT Support Agent, Employee)
Description	NVARCHAR(255)	NULL	Role description

2. User

Purpose: Stores information about every user in the system.

Column	Data Type	Constraints	Description
Id	INT	PK	Unique user identifier
RoleId	INT	FK → Role(Id), NOT NULL	User role
FirstName	NVARCHAR(50)	NOT NULL	User first name
LastName	NVARCHAR(50)	NOT NULL	User last name
Email	NVARCHAR(100)	UNIQUE, NOT NULL	User email address
PasswordHash	NVARCHAR(255)	NOT NULL	Encrypted password
PhoneNumber	NVARCHAR(20)	NULL	User phone number
ProfileImageUrl	NVARCHAR(255)	NULL	Profile image path or URL
IsActive	BIT	NOT NULL	Indicates whether the account is active
CreatedDate	DATETIME	NOT NULL	Account creation date
LastLoginDate	DATETIME	NULL	Last successful login

3. Category

Purpose: Stores ticket and knowledge base categories.

Column	Data Type	Constraints	Description
Id	INT	PK	Unique category identifier
Name	NVARCHAR(50)	NOT NULL	Category name
Description	NVARCHAR(255)	NULL	Category description

4. Priority

Purpose: Stores ticket priority levels.

Column	Data Type	Constraints	Description
Id	INT	PK	Unique priority identifier
Name	NVARCHAR(30)	NOT NULL	Priority name
DisplayOrder	INT	NOT NULL	Display order

5. Status

Purpose: Stores ticket status values.

Column	Data Type	Constraints	Description
Id	INT	PK	Unique status identifier
Name	NVARCHAR(30)	NOT NULL	Status name
DisplayOrder	INT	NOT NULL	Display order

6. Ticket

Purpose: Stores all support tickets submitted by employees.

Column	Data Type	Constraints	Description
Id	INT	PK	Unique ticket identifier

Column	Data Type	Constraints	Description
ReferenceNumber	NVARCHAR(20)	UNIQUE, NOT NULL	Ticket reference number
Title	NVARCHAR(150)	NOT NULL	Ticket title
Description	NVARCHAR(MAX)	NOT NULL	Ticket description
CategoryId	INT	FK → Category(Id), NOT NULL	Ticket category
PriorityId	INT	FK → Priority(Id), NOT NULL	Ticket priority
StatusId	INT	FK → Status(Id), NOT NULL	Current ticket status
CreatedByUserId	INT	FK → User(Id), NOT NULL	Employee who created the ticket
AssignedToUserId	INT	FK → User(Id), NULL	Assigned IT support agent
Solution	NVARCHAR(MAX)	NULL	Final solution or resolution
CreatedDate	DATETIME	NOT NULL	Ticket creation date
DueDate	DATETIME	NULL	Expected completion date
ResolvedDate	DATETIME	NULL	Resolution date
ClosedDate	DATETIME	NULL	Closing date

7. TicketComment

Purpose: Stores comments and internal notes for tickets.

Column	Data Type	Constraints	Description
Id	INT	PK	Unique comment identifier

Column	Data Type	Constraints	Description
TicketId	INT	FK → Ticket(Id), NOT NULL	Related ticket
UserId	INT	FK → User(Id), NOT NULL	Comment author
Comment	NVARCHAR(MAX)	NOT NULL	Comment text
IsInternal	BIT	NOT NULL	Indicates whether the comment is internal
CreatedDate	DATETIME	NOT NULL	Comment creation date

8. TicketAttachment

Purpose: Stores files attached to tickets.

Column	Data Type	Constraints	Description
Id	INT	PK	Unique attachment identifier
TicketId	INT	FK → Ticket(Id), NOT NULL	Related ticket
UserId	INT	FK → User(Id), NOT NULL	User who uploaded the file
FileName	NVARCHAR(255)	NOT NULL	File name
FilePath	NVARCHAR(500)	NOT NULL	Storage path or URL
FileSize	BIGINT	NOT NULL	File size in bytes
ContentType	NVARCHAR(100)	NOT NULL	File MIME type
UploadedDate	DATETIME	NOT NULL	Upload date

9. Notification

Purpose: Stores system notifications for users.

Column	Data Type	Constraints	Description
Id	INT	PK	Unique notification identifier
UserId	INT	FK → User(Id), NOT NULL	Notification recipient
Title	NVARCHAR(100)	NOT NULL	Notification title
Message	NVARCHAR(MAX)	NOT NULL	Notification message
IsRead	BIT	NOT NULL	Indicates whether the notification has been read
CreatedDate	DATETIME	NOT NULL	Notification creation date

10. ActivityLog

Purpose: Stores audit records for important user and ticket activities.

Column	Data Type	Constraints	Description
Id	INT	PK	Unique activity log identifier
UserId	INT	FK → User(Id), NOT NULL	User who performed the action
TicketId	INT	FK → Ticket(Id), NULL	Related ticket (if applicable)
Action	NVARCHAR(100)	NOT NULL	Action performed
Description	NVARCHAR(MAX)	NULL	Additional action details
CreatedDate	DATETIME	NOT NULL	Date and time of the activity

11. TicketAssignment

Purpose: Stores ticket assignment and reassignment history.

Column	Data Type	Constraints	Description
Id	INT	PK	Unique assignment identifier

Column	Data Type	Constraints	Description
TicketId	INT	FK → Ticket(Id), NOT NULL	Assigned ticket
AssignedToUserId	INT	FK → User(Id), NOT NULL	Assigned IT support agent
AssignedByUserId	INT	FK → User(Id), NOT NULL	User who performed the assignment
AssignedDate	DATETIME	NOT NULL	Assignment date

12. KnowledgeBaseArticle

Purpose: Stores troubleshooting guides, FAQs, and help articles.

Column	Data Type	Constraints	Description
Id	INT	PK	Unique article identifier
Title	NVARCHAR(200)	NOT NULL	Article title
Summary	NVARCHAR(500)	NULL	Short article summary
Content	NVARCHAR(MAX)	NOT NULL	Full article content
CategoryId	INT	FK → Category(Id), NOT NULL	Related category
CreatedByUserId	INT	FK → User(Id), NOT NULL	Article author
ApprovedByUserId	INT	FK → User(Id), NULL	User who approved the article
IsApproved	BIT	NOT NULL	Approval status
CreatedDate	DATETIME	NOT NULL	Article creation date
PublishedDate	DATETIME	NULL	Publication date